

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12408244
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Nakayama; Masaaki

Light-emitting element driving circuit

Abstract

A light-emitting element driving circuit (1) includes: a first buffer (6) provided between the transmission terminal (1B) and the communication circuit (2), with the output side of the first buffer connected to the transmission terminal and the input side of the first buffer connected to the communication circuit; and a determination circuit (32) configured to determine whether the controller (15) is connected or not based on a first signal (S1) at the output terminal of the first buffer and a second signal (S2) at the input terminal of the first buffer. If the determination circuit determines that the controller is not connected, the communication circuit is disabled from processing the signal received via the reception terminal (1A), and the dimming signal generator (31) generates the dimming signal (Spwm) based on a pulse signal (Sp) fed to the reception terminal or the transmission terminal.

Inventors:	Nakayama; Masaaki (Kyoto, JP)
Applicant:	ROHM CO., LTD. (Kyoto, JP)
Family ID:	83396944
Assignee:	Rohm Co., Ltd. (Kyoto, JP)
Appl. No.:	18/469880
Filed:	September 19, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240015864 A1	Jan. 11, 2024

Foreign Application Priority Data

JP	2021-049795	Mar. 24, 2021
----	-------------	---------------

Related U.S. Application Data

continuation parent-doc WO PCT/JP2022/009095 20220303 PENDING child-doc US 18469880

Publication Classification

Int. Cl.: H05B45/10 (20200101); H05B45/325 (20200101)

U.S. Cl.:

CPC H05B45/10 (20200101); H05B45/325 (20200101);

Field of Classification Search

CPC: H05B (45/10); H05B (45/325); H05B (45/345); Y02B (20/40)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2017/0086281	12/2016	Avrahamy	N/A	H05B 39/086
2018/0042064	12/2017	Norton	N/A	H04W 4/70
2019/0335551	12/2018	Williams	N/A	H05B 45/24

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2012182010	12/2011	JP	N/A
2017-028866	12/2016	JP	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion in International Appln. No. PCT/JP2022/009095, mailed on May 10, 2022, 12 pages (with machine translation). cited by applicant
JP OA—Japanese Patent Office, Office Action in Japanese Appln. No. 2023-508885, dated Jul. 22, 2025, 6 pages (with English translation) . cited by applicant

Primary Examiner: Riyami; Abdullah A

Assistant Examiner: Kaiser; Syed M

Attorney, Agent or Firm: Fish & Richardson P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This nonprovisional application is a continuation application of International Patent Application No. PCT/JP2022/009095 filed on Mar. 3, 2022, which claims priority Japanese Patent Application No. 2021-049795 filed on Mar. 24, 2021, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

(1) The present disclosure relates to a light-emitting element driving circuit.

BACKGROUND ART

(2) Various light-emitting element driving circuits are known for driving light-emitting elements such as LEDs (light emitting diodes).

(3) Conventional LED driving circuits include a current driver for passing a current through an LED. Some conventional LED driving circuits perform PWM (pulse width modulation) control on the current driver based on communication with an external processor (see, for example, Patent Document 1). In PWM control, the current driver is turned on and off based on a PWM signal. This permits PWM dimming of the LED.

CITATION LIST

Patent Literature

(4) Patent Document 1: Japanese Unexamined Patent Application Publication No. 2017-28866

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram showing the schematic block configuration of an LED driving circuit according to a first embodiment.

(2) FIG. 2 is a diagram showing the LED driving circuit according to the first embodiment used with an MCU connected to it.

(3) FIG. 3 is a diagram showing the LED driving circuit according to the first embodiment used with no MCU connected to it.

(4) FIG. 4 is a diagram showing the schematic block configuration of an LED driving circuit according to a second embodiment.

(5) FIG. 5 is a diagram showing the LED driving circuit according to the second embodiment used with an MCU connected to it.

(6) FIG. 6 is a diagram showing the LED driving circuit according to the second embodiment used with no MCU connected to it.

(7) FIG. 7 is a diagram showing the schematic block configuration of an LED driving circuit according to a comparative example.

DESCRIPTION OF EMBODIMENTS

1. Comparative Example

(8) First, before a description of illustrative embodiments of the present invention, a comparative example will be described. FIG. 7 is a diagram showing the schematic block configuration of an LED driving circuit 10 according to the comparative example. The LED driving circuit 10 shown in FIG. 7 includes a UART 11, a logic circuit 12, and a constant-current circuit 13. The LED driving circuit 10 has an RX terminal 10A, a TX terminal 10B, an LED terminal 10C, and a ground terminal 10D as external terminals for establishing electrical connections with the outside.

(9) The UART (universal asynchronous receiver/transmitter) 11 is a circuit that performs two-way conversion between serial and parallel signals. Outside the LED driving circuit 10, an MCU (microcontroller unit) 15 is provided. The MCU 15 is connectable to the RX and TX terminals 10A and 10B. The UART 11 receives a reception signal Srx from the MCU 15 via the RX terminal 10A. The UART 11 transmits a transmission signal Stx to the MCU 15 via the TX terminal 10B. That is, the UART 11 is a communication circuit configured to be able to communicate with the MCU 15.

(10) The constant-current circuit 13 is a circuit for passing a constant LED current ILED in an LED 20 provided outside the LED driving circuit 10. The anode of the LED 20 is connected to an application terminal for a supply voltage Vcc. The cathode of the LED 20 is connected to the LED terminal 10C. The ground terminal 10D is connected to an application terminal for a ground

potential. The constant-current circuit **13** is provided to connect between the LED and ground terminals **10C** and **10D**. The LED **20** is not limited to a configuration in which a plurality of LED elements are connected in series as shown in FIG. 7, and may be, for example, a single LED element.

(11) The UART **11** processes the reception signal Srx from the MCU **15** and gives a command to the logic circuit **12**. The logic circuit **12** receives the command and generates a PWM signal Spwm. The PWM signal Spwm is a pulse signal having levels corresponding to an on period and an off period respectively. The constant-current circuit **13** is turned on and off based on the PWM signal Spwm. When the constant-current circuit **13** is on, the LED current ILED flows, and when the constant-current circuit **13** is off, the LED current ILED does not flow. Thus, according to the on duty of the PWM signal Spwm (that is, the proportion of the on period in the pulse period), PWM dimming of the LED **20** can be performed.

(12) However, with this LED driving circuit **10**, if the MCU **15** is not used, communication with the MCU **15** cannot be performed, so light emission control for the LED **20** cannot be performed. That is, the use of the MCU **15** is prerequisite and the LED driving circuit **10** cannot be used stand-alone.

(13) To cope with that, additional terminals may be provided to allow the LED driving circuit **10** to be used stand-alone; however, increasing the number of terminals in the LED driving circuit **10** leads to an increase in the number of terminals in the IC (integrated circuit) package that results from modularizing the LED driving circuit **10**, which is undesirable.

2. First Embodiment

(14) <2-1. Configuration of the LED Driving Circuit>

(15) Hereinafter, various embodiments devised to solve the above problems will be described. FIG. 1 is a diagram showing the schematic block configuration of an LED driving circuit **1** according to a first embodiment.

(16) The LED driving circuit **1** shown in FIG. 1 includes a UART **2** (communication circuit), a logic circuit **3**, a constant-current circuit **4**, and buffers **5** and **6**. The LED driving circuit **1** has an RX terminal **1A** (reception terminal), a TX terminal **1B** (transmission terminal), an LED terminal **1C**, and a ground terminal **1D** for establishing electrical connections with the outside.

(17) The RX and TX terminals **1A** and **1B** can be connected to an external MCU (controller) not shown in FIG. 1 (see FIG. 2, which will be referred to later). The RX terminal **1A** is connected to the UART **2** via the buffer **5**. The input side of the buffer **5** is its side connected to the RX terminal **1A** and the output side of the buffer **5** is its side connected to the UART **2**. The UART **2** can receive a reception signal from the MCU via the RX terminal **1A** and the buffer **5**.

(18) The TX terminal **1B** is connected to the UART **2** via the buffer **6**. The input side of the buffer **6** is its side connected to the UART **2** and the output side of the buffer **6** is its side connected to the TX terminal **1B**. The UART **2** can transmit a transmission signal to the MCU via the buffer **6** and the TX terminal **1B**.

(19) The logic circuit **3** has a PWM signal generator **31** (dimming controller) and a determination circuit **32**.

(20) The PWM signal generator **31** is configured to be able to generate a PWM signal Spwm (dimming signal) according to a command from the UART **2**. The PWM signal generator **31** is connected to the output terminal of the buffer **5**. Thus, in a stand-alone mode, the PWM signal generator **31** generates the PWM signal Spwm based on a signal fed from the buffer **5**. The stand-alone mode is a mode in which the LED driving circuit **1** is used without the LED driving circuit **1** being connected to the MCU.

(21) The determination circuit **32** is configured to be able to monitor a first signal S1 at the output terminal of the buffer **6** (the signal at a node N1) and a second signal S2 at the input terminal of the buffer **6** (the signal at a node N2). Based on the first and second signals S1 and S2, the determination circuit **32** determines whether the LED driving circuit **1** is connected to the MCU. If

the determination circuit **32** determines that the MCU is unconnected, it shifts the LED driving circuit **1** to the stand-alone mode.

(22) Incidentally, between the node **N1** and the determination circuit **32**, a buffer may be provided with its input side connected to the node **N1** and its output side connected to the determination circuit **32**.

(23) The interconnections around the constant-current circuit **4** and the LED **20** are the same as in the comparative example described above. The constant-current circuit **4** is provided to connect between the LED and ground terminals **1C** and **1D**. The constant-current circuit **4** is turned on and off based on the PWM signal **Spwm**.

(24) <2-2. Normal Mode>

(25) The operation of the LED driving circuit **1** configured as described above will be described. FIG. **2** is a diagram showing the LED driving circuit **1** used with the MCU **15** connected to the LED driving circuit **1**. That is, FIG. **2** is a diagram showing the LED driving circuit **1** used in a normal mode.

(26) As shown in FIG. **2**, owing to the MCU **15** being connected to the RX and TX terminals **1A** and **1B**, the UART **2** can receive the reception signal **Srx** from the MCU **15** via the RX terminal **1A** and the buffer **5**, and can transmit the transmission signal **Stx** to the MCU **15** via the buffer **6** and the TX terminal **1B**.

(27) The reception signal **Srx** received by the UART **2** from the MCU **15** and the transmission signal **Stx** transmitted from the UART **2** to the MCU **15** are each a signal that can have either high or low level. Initially, the UART **2** receives the reception signal **Srx** and then transmits the transmission signal **Stx**. Meanwhile, the determination circuit **32** monitors the first and second signals **S1** and **S2**, and if it senses that the first signal **S1** is at high level when the second signal **S2** is being fed from the UART **2**, it determines that the MCU **15** is connected to the LED driving circuit **1**. As a result, the determination circuit **32** shifts the LED driving circuit **1** to the normal mode.

(28) At that time, the determination circuit **32** enables the UART **2** to process the reception signal **Srx**. Thus, the UART **2** gives a command to the PWM signal generator **31** based on the processing of the reception signal **Srx**. In response to the command, the PWM signal generator **31** generates the PWM signal **Spwm**. The constant-current circuit **4** is turned on and off based on the PWM signal **Spwm**. Thus, in the normal mode, based on the reception signal **Srx** from the MCU **15**, PWM dimming of the LED **20** is performed.

(29) <2-3. Stand-Alone Mode>

(30) FIG. **3** is a diagram showing the LED driving circuit **1** used without the MCU **15** connected to the LED driving circuit **1**. That is, FIG. **3** is a diagram showing the LED driving circuit **1** used in a stand-alone mode.

(31) As shown in FIG. **3**, when the LED driving circuit **1** is used in the stand-alone mode, the RX and TX terminals **1A** and **1B** are short-circuited together outside the LED driving device **1** and the RX and TX terminals **1A** and **1B** are fed with a pulse signal **Sp**.

(32) In this case, initially, the determination circuit **32** monitors the first and second signals **S1** and **S2**. The determination circuit **32** senses that the first signal **S1** is at high level due to the pulse signal **Sp** when the second signal **S2** is not being fed, and determines that the MCU is not connected to the LED driving circuit **1**. As a result, the determination circuit **32** shifts the LED driving circuit **1** to the stand-alone mode.

(33) At that time, the determination circuit **32** disables the UART **2** from processing the signal received from the RX terminal **1A** (here, the pulse signal **Sp**). Also, the determination circuit **32** commands the PWM signal generator **31** to operate in the stand-alone mode. In response to the command, the PWM signal generator **31** generates the PWM signal **Spwm** based on the signal fed from the buffer **5**. At this time, the PWM signal **Spwm** is generated by through processing. That is, the PWM signal **Spwm** is at high level during the high level period of the pulse signal **Sp** and is at

low level during the low level period of the pulse signal Sp.

(34) The constant-current circuit **4** is turned on and off based on the PWM signal Spwm. Thus, in the stand-alone mode, PWM dimming of the LED **20** is performed based on the pulse signal Sp.

(35) As described above, according to this embodiment, if the MCU **15** is not used, the LED driving circuit **1** can be used with the RX and TX terminals **1A** and **1B** short-circuited together and fed with the pulse signal Sp, so that it operates stand-alone to perform light emission control for the LED **20**. This eliminates the need to add a new terminal in the LED driving circuit **1** and helps avoid an increase in the number of terminals.

3. Second Embodiment

(36) FIG. **4** is a diagram showing the schematic block configuration of an LED driving circuit **1X** according to a second embodiment. Described below will be differences in configuration of the LED driving circuit **1X** shown in FIG. **4** from the first embodiment (see FIG. **1**).

(37) The LED driving circuit **1X** has a buffer **7**. The input terminal of the buffer **7** is connected to a node N**1**. The output terminal of the buffer **7** is connected to a determination circuit **32**. In the LED driving circuit **1X**, as opposed to the first embodiment, the PWM signal generator **31** is connected to the output terminal of the buffer **7** instead of to the output terminal of the buffer **5**.

(38) The operation of the LED driving circuit **1X** configured as described above will be described. FIG. **5** is a diagram showing the LED driving circuit **1X** used with an MCU **15** connected to the LED driving circuit **1X**. That is, FIG. **5** is a diagram showing the LED driving circuit **1X** used in a normal mode.

(39) Initially, the UART **2** receives a reception signal Srx and then transmits a transmission signal Stx. Meanwhile, the determination circuit **32** monitors a first and second signal S**1** and S**2**. The determination circuit **32** monitors the first signal S**1** via the buffer **7**. If the determination circuit **32** senses that the first signal S**1** is at high level when the second signal S**2** is being fed from the UART **2**, it determines that the MCU **15** is connected to the LED driving circuit **1X**. As a result, the determination circuit **32** shifts the LED driving circuit **1X** to the normal mode. The subsequent operation is similar to that in the first embodiment. Thus, the LED driving circuit **1X** can operate in the normal mode while being connected to the MCU **15**.

(40) FIG. **6** is a diagram showing the LED driving circuit **1X** used without the MCU **15** connected to the LED driving circuit **1X**. That is, FIG. **6** is a diagram showing the LED driving circuit **1X** used in a stand-alone mode.

(41) As shown in FIG. **6**, if the MCU **15** is not connected, an RX terminal **1A** is open and a TX terminal **1B** is fed with a pulse signal Sp from the outside. Initially the determination circuit **32** monitors the first and second signals S**1** and S**2**. The determination circuit **32** senses that the first signal S**1** is at high level due to the pulse signal Sp when the second signal S**2** is not being fed, and determines that the MCU is not connected to the LED driving circuit **1X**. As a result, the determination circuit **32** shifts the LED driving circuit **1X** to the stand-alone mode.

(42) At that time, the determination circuit **32** disables the UART **2** from processing the signal received from the RX terminal **1A**. Also, the determination circuit **32** commands the PWM signal generator **31** to operate in the stand-alone mode. In response to the command, the PWM signal generator **31** generates the PWM signal Spwm based on the pulse signal Sp fed via the TX terminal **1B** and the buffer **7**. At this time, the PWM signal Spwm is generated by through processing as in the first embodiment.

(43) Thus, the LED driving circuit **1X** according to this embodiment, like that of the first embodiment, can operate in the stand-alone mode to perform light emission control for the LED **20**. Also, as in the first embodiment, no additional terminals are required.

(44) Here, using a buffer **7** with fast response makes it possible to transmit the pulse signal Sp to the PWM signal generator **31** with good response and to improve the characteristics of an LED current ILED. However, considering that the buffer **5** used in the normal mode (see FIG. **5**) also needs to have fast response, in the first embodiment (see FIG. **1**), a fast-response buffer used for

generating the PWM signal Spwm can be shared as the buffer 5 in the normal mode and the stand-alone mode.

4. IC Package

(45) The LED **20** and the LED driving circuit **1** (or **1X**) are, for example, built into one package. In that case, the IC package is configured with an IC chip having the LED **20** integrated in it and an IC chip having the LED driving circuit **1** (or **1X**) integrated in it that are sealed in with a sealing material (resin or the like). The IC package also has terminals (lead frames) for establishing electrical connection with the outside. The IC chip having the LED **20** integrated in it, the IC chip having the LED driving circuit **1** (or **1X**) integrated in it, and the terminals are interconnected by bonding wires or the like.

(46) As described above, the LED driving circuit **1** (or **1X**) does not require additional terminals. Thus, for an IC package configured as described above, with which it is difficult to increase the number of terminals because of the standard, the LED driving circuit **1** (or **1X**) is particularly effective.

(47) The LED and the LED driving circuit may be built into different packages.

5. Other Modifications

(48) While illustrative embodiments have been described above, those embodiments allow for various modifications made without departure from the technical scope of the present invention.

(49) For example, the communication circuit provided in the LED driving circuit for communication with the MCU is not limited to a UART, but may be one that performs communication complying with SPI (serial peripheral interface) or I2C (inter-integrated circuit).

6. Notes

(50) As described above, for example, according to one aspect of the present disclosure, a light-emitting element driving circuit (**1**) includes: a reception terminal (**1A**) and a transmission terminal (**1B**) connectable to an external controller (**15**); a communication circuit (**2**) configured to receive a reception signal from the controller via the reception terminal and transmit a transmission signal to the controller via the transmission terminal; a dimming signal generator (**31**) configured to generate a dimming signal (Spwm); a constant-current circuit (**4**) configured to pass a current in a light-emitting element (**20**) and is turned on and off based on the dimming signal; a first buffer (**6**) provided between the transmission terminal and the communication circuit, with the output side of the first buffer connected to the transmission terminal and the input side of the first buffer connected to the communication circuit; and a determination circuit (**32**) configured to determine whether the controller is connected or not based on a first signal (S1) at the output terminal of the first buffer and a second signal (S2) at the input terminal of the first buffer, wherein if the determination circuit determines that the controller is connected, the communication circuit is enabled to process a signal received via the reception terminal, and the dimming signal generator generates the dimming signal based on a command from the communication circuit, and if the determination circuit determines that the controller is not connected, the communication circuit is disabled from processing the signal received via the reception terminal, and the dimming signal generator generates the dimming signal based on a pulse signal (Sp) fed to the reception terminal or the transmission terminal. (A first configuration.)

(51) In the light-emitting element driving circuit of the first configuration described above, there may be further provided a second buffer (**5**) between the reception terminal and the communication circuit, with the input side of the second buffer connected to the reception terminal and the output side of the second buffer connected to the communication circuit. The output terminal of the second buffer may be connected to the dimming signal generator, and if the controller is not used, the reception and transmission terminals may be short-circuited together. (A second configuration.)

(52) In the light-emitting element driving circuit of the second configuration described above, there may be further provided a third buffer between a node (N1) to which the transmission terminal and the output terminal of the first buffer are connected and the determination circuit, with the input

side of the third buffer connected to the node and the output side of the third buffer connected to the determination circuit. (A third configuration.)

(53) In the light-emitting element driving circuit of the first configuration described above, there may be further provided a fourth buffer (7) between a node (N1) to which the transmission terminal and the output terminal of the first buffer are connected and the determination circuit, with the input side of the fourth buffer connected to the node and the output side of the fourth buffer connected to the determination circuit. The output terminal of the fourth buffer may be connected to the dimming signal generator. (A fourth configuration.)

(54) In the light-emitting element driving circuit of any one of the first to fourth configuration described above, the dimming signal may be a PWM signal. (A fifth configuration.)

(55) In the light-emitting element driving circuit of any one of the first to fifth configuration described above, the communication circuit may be a UART (universal asynchronous receiver/transmitter). (A sixth configuration.)

(56) In the light-emitting element driving circuit of any one of the first to sixth configuration described above, the controller may be an MCU (microcontroller unit). (A seventh configuration.)

(57) In the light-emitting element driving circuit of any one of the first to seventh configuration described above, the light-emitting element may be an LED (light-emitting diode). (An eighth configuration.)

(58) According to another aspect of what is disclosed herein, an IC package includes the light-emitting element driving circuit according to the eighth configuration described above and the LED are built into one package.

INDUSTRIAL APPLICABILITY

(59) The invention disclosed herein finds applications in, for example, LED driving circuits for various uses.

REFERENCE SIGNS LIST

(60) **1**, **1X** LED driving circuit **1A** RX terminal **1B** TX terminal **1C** LED terminal **1D** ground terminal **2** UART **3** logic circuit **4** constant-current circuit **5**, **6**, **7** buffer **10** LED driving circuit **10A** RX terminal **10B** TX terminal **10C** LED terminal **10D** ground terminal **11** UART **12** logic circuit **13** constant-current circuit **20** LED **31** PWM signal generator **32** determination circuit

Claims

1. A light-emitting element driving circuit comprising: a reception terminal and a transmission terminal connectable to an external controller; a communication circuit configured to receive a reception signal from the controller via the reception terminal and transmit a transmission signal to the controller via the transmission terminal; a dimming signal generator configured to generate a dimming signal; a constant-current circuit configured to pass a current in a light-emitting element and be turned on and off based on the dimming signal; a first buffer provided between the transmission terminal and the communication circuit, with an output side of the first buffer connected to the transmission terminal and an input side of the first buffer connected to the communication circuit; and a determination circuit configured to determine whether the controller is connected or not based on a first signal at the output terminal of the first buffer and a second signal at the input terminal of the first buffer, wherein if the determination circuit determines that the controller is connected, the communication circuit is enabled to process a signal received via the reception terminal, and the dimming signal generator generates the dimming signal based on a command from the communication circuit, and if the determination circuit determines that the controller is not connected, the communication circuit is disabled from processing the signal received via the reception terminal, and the dimming signal generator generates the dimming signal based on a pulse signal fed to the reception terminal or the transmission terminal.

2. The light-emitting element driving circuit according to claim 1, further comprising: a second

buffer provided between the reception terminal and the communication circuit, with an input side of the second buffer connected to the reception terminal and an output side of the second buffer connected to the communication circuit, wherein the output terminal of the second buffer is connected to the dimming signal generator, and if the controller is not used, the reception and transmission terminals are short-circuited together.

3. The light-emitting element driving circuit according to claim 2, further comprising: a third buffer provided between a node to which the transmission terminal and the output terminal of the first buffer are connected and the determination circuit, with an input side of the third buffer connected to the node and an output side of the third buffer connected to the determination circuit.

4. The light-emitting element driving circuit according to claim 1, further comprising: a fourth buffer provided between a node to which the transmission terminal and the output terminal of the first buffer are connected and the determination circuit, with an input side of the fourth buffer connected to the node and an output side of the fourth buffer connected to the determination circuit, wherein the output terminal of the fourth buffer is connected to the dimming signal generator.

5. The light-emitting element driving circuit according to claim 1, wherein the dimming signal is a PWM signal.

6. The light-emitting element driving circuit according to claim 1, wherein the communication circuit is a UART (universal asynchronous receiver/transmitter).

7. The light-emitting element driving circuit according to claim 1, wherein the controller is an MCU (microcontroller unit).

8. The light-emitting element driving circuit according to claim 1, wherein the light-emitting element is an LED (light-emitting diode).

9. An IC package having the light-emitting element driving circuit according to claim 8 and the LED built into one package.
